

Premature Deindustrialization

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Introduction: The Power and Decline of Manufacturing

- **Historical Engine of Growth:** Industrialization created the modern world. It drove initial divergence between rich and poor nations, and later permitted rapid catch-up for late developers.
- **The Post-Industrial Phase:** Advanced economies have been deindustrializing for decades. This is expected and is most noticeable in employment rather than real output.
- **The Unnoticed Trend:** Low- and middle-income countries (outside of Asia) have been experiencing a much more striking and puzzling decline in manufacturing since the 1980s.
- **The Core Thesis:** Developing countries are deindustrializing without ever having gone through a proper experience of industrialization.
- **The Stakes:** Historically, it was industrialization that allowed poor nations to achieve rapid economic growth.

Table 1: Indicators of Global Manufacturing Activity

Table: In 2005 constant USD

Year	World	United States	Western Europe	Latin America & Caribbean	Asia (excluding China)	China	Sub-Saharan Africa	Other
Shares in global MVA								
1970	1.00	0.26	0.24	0.06	0.15	0.00	0.01	0.27
1980	1.00	0.22	0.21	0.08	0.18	0.01	0.01	0.29
1990	1.00	0.21	0.19	0.06	0.24	0.02	0.01	0.26
2000	1.00	0.24	0.16	0.07	0.24	0.06	0.01	0.22
2010	1.00	0.20	0.13	0.06	0.26	0.16	0.01	0.18
2013	1.00	0.19	0.13	0.06	0.26	0.18	0.01	0.17
MVA share in GDP								
1970	0.17	0.13	0.22	0.20	0.16	0.09	0.14	
1980	0.16	0.12	0.20	0.20	0.16	0.16	0.15	
1990	0.16	0.12	0.19	0.19	0.19	0.18	0.15	
2000	0.17	0.13	0.18	0.19	0.19	0.29	0.13	
2010	0.18	0.13	0.18	0.17	0.21	0.36	0.11	
2013	0.18	0.13	0.18	0.16	0.20	0.36	0.11	

Measuring Deindustrialization

Three measures used in the paper:

- ① **manemp**: Manufacturing employment share (strongest decline)
 - ② **nommva**: Manufacturing value-added at current prices (moderate decline)
 - ③ **realmva**: Manufacturing value-added at constant prices (weaker decline)
- Manufacturing employment has fallen 11.7 percentage points since the 1950s.
 - Real manufacturing output shares remain more robust in advanced economies.
 - Stripping away price changes (using *realmva*) is crucial to understand physical output.

Deindustrialization Over Time

The Baseline Regression:

$$\text{share}_{it} = \beta_0 + \beta_1 \ln \text{pop}_{it} + \cdots + \beta_3 \ln y_{it} + \cdots + \sum_i \gamma_i D_i + \sum_T \phi_T \text{PER}_T + \epsilon_{it}$$

Isolating decade dummies (PER_T) while controlling for income (y) and country fixed effects (D_i).

Purpose:

- Proves deindustrialization is a **global secular trend**, happening independently of any single country's income level.

Key Results (1950s → 2000s):

- **Employment** (manemp): Clear fall ($\downarrow$ 11.7 percentage points).
- **Nominal Output** (nommva): Also a decline ($\downarrow$ 8.5 percentage points).
- **Real Output** (realmva): Smaller decline (depending on sample type: $\downarrow$ 3.5 – 5.9 percentage points).

Takeaway: Global physical manufacturing volume has not collapsed; modern factories simply produce the similar output using far fewer workers.

Deindustrialization in different country groups

Bunch of regressions for groups of countries:

Developed, Latin America, Asian, Sub-Saharan Africa (with and without Mauritius)

Major regional patterns:

- **Advanced economies:** Big losses in *manemp* and *nommva* but fairly stable *realmva*
- **Asia (manufactures exporters):** *Manemp* and *realmva* are positive.
- **Latin America:** Hardest hit strong losses in both employment and real output (nominal not as pronounced).
- **Sub-Saharan Africa (without Mauritius):** Dramatic output deindustrialization (in all three measures) despite still being very poor.

Conclusion: The brunt of globalization's impact has been borne by developing countries, not advanced economies.

Who Bears the Costs? Employment by Skill

- Between 1995–2009, manufacturing's share of employment fell by 4 percentage points.
- The entire decline came from low-skill workers.**
- Medium-skill was stable; high-skill slightly increased.
- Implication:* Unskilled workers are displaced into low-productivity services.

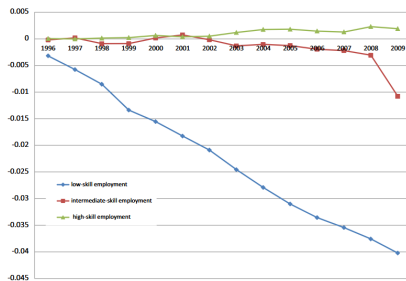


Fig. 4 Estimated year coefficients for employment of different skill types

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The Core Puzzle: Shifting Curves

- Before 1990: Employment share peaked at $\sim \$11,000$ per capita.
- After 1990: Employment share peaks at $\sim \$4,300$ per capita.
- **Result:** The hump-shaped curve has moved closer to the origin (a $\sim 60\%$ reduction in peak income).

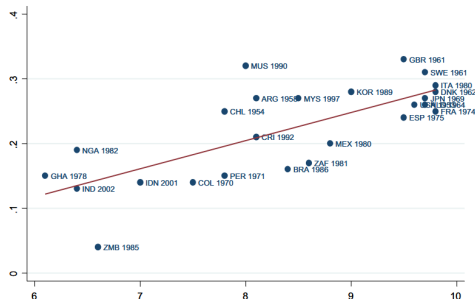


Fig 5: Simulated manufacturing employment shares (pre/post 1990)

The Analytical Framework: Model Setup

A Two-Sector Model: Manufacturing (m) and Non-manufacturing (n).
Let α be the share of employment in manufacturing.

Production :

$$q_m^s = \theta_m \alpha^{\beta_m}$$

$$q_n^s = \theta_n (1 - \alpha)^{\beta_n}$$

Demand (σ = elasticity of substitution; hats denote rates of change):

$$\hat{q}_m^d - \hat{q}_n^d = -\sigma(\hat{p}_m - \hat{p}_n)$$

Labor Market Equilibrium (equating value marginal product):

$$\beta_m p_m \theta_m \alpha^{\beta_m - 1} = \beta_n p_n \theta_n (1 - \alpha)^{\beta_n - 1}$$

The Model: Large “Closed” Economy (Advanced)

- Prices are determined domestically.
- Comparative statics for manufacturing employment share ($d\alpha$):

$$d\alpha = \psi \left[\left(\frac{\sigma - \lambda}{\sigma} \right) \hat{\theta}_m - \left(\frac{\sigma - 1}{\sigma} \right) \hat{\theta}_n + \frac{1}{\sigma} \frac{dx}{q_m^d} \right]$$

where

$$\psi = \left[\frac{1}{\alpha(1 - \beta_m)} + \frac{1}{(1 - \alpha)(1 - \beta_n)} + \frac{1}{\sigma} \left(\frac{\lambda}{\alpha\beta_m} + \frac{1}{(1 - \alpha)\beta_n} \right) \right]^{-1} > 0$$

and

$$\lambda = \frac{q_m^s}{q_m^d}$$

Analytical Framework: Summary of Results

Small Open Economy (Developing) Comparative Statics:

$$d\alpha = \left[\frac{1}{\alpha(1 - \beta_m)} + \frac{1}{(1 - \alpha)(1 - \beta_n)} \right]^{-1} \left[\hat{p}_m + \hat{\theta}_m - \hat{\theta}_n \right]$$

Effects of Shocks on Measures of Industrialization:

A. "Closed" economy (with $\sigma < 1$)

Effect on:	Technology shock: $\hat{\theta}_m - \hat{\theta}_n > 0$	Trade shock: $dx < 0$	Adv. Domestic Demand
$manemp (d\alpha)$	—	—	—
$realmva (d\alpha_q)$	+	—	—

B. Small open economy

Effect on:	Technology shock: $\hat{\theta}_m - \hat{\theta}_n > 0$	Ext. Price shock: $\hat{p}_m < 0$	Adv. Domestic Demand
$manemp (d\alpha)$	+	—	0
$realmva (d\alpha_q)$	+	—	0

Technology vs. Trade: Importing Deindustrialization

Advanced economies: Technology is the primary culprit.

- Rapid productivity growth reduces labor demand.
- Employment declines despite stable physical output.

Developing countries: Trade and globalization dominate.

- Global TFP growth + Asian manufacturing dominance caused world prices of manufactures to plummet ($\hat{p}_m < 0$).
- Developing nations opened up to trade and were exposed to this price collapse.
- Without a strong comparative advantage, they “imported” deindustrialization.

Why This Matters for Growth

Manufacturing is the “escalator” for developing economies because:

- ① It absorbs large quantities of unskilled labor.
- ② It exhibits **unconditional labor productivity convergence** (unlike services).
- ③ It is tradable — not constrained by home-market demand or local purchasing power.
- ④ It generates more rapid productivity growth long-term.

The problem: Without manufacturing, developing countries lose their main avenue for rapid economic convergence.

Already visible in the data:

- **Latin America:** As manufacturing shrunk, informality and low-productivity services grew.
- **Africa:** Urban migrants crowding into petty services instead of factories.
- Structural change has become *growth-reducing* in these regions (unlike the 1950–1975 era).

Services cannot easily replace manufacturing:

- High-productivity services (IT, finance) are highly skill-intensive.
- Low-productivity services don't provide a growth engine.
- Non-tradable services face strict home-market constraints.

Historical role of industrialization:

- Created organized labor movements.
- Labor bargained for franchise expansion and the welfare state.
- Class-based politics enabled modern liberal democracy.

The concern for developing nations:

- Weak organized labor due to premature deindustrialization.
- May foster different political paths.
- Identity/ethnic cleavages may substitute for class cleavages.
- Higher risk of “electoral” or “illiberal” democracies.

What Can Developing Countries Do?

Modest growth is still possible through “fundamentals”:

- Improved institutions and governance.
- Growing human capital and skills.

But rapid catch-up is historically difficult without manufacturing:

- Advanced economies manage 1.5–2% growth in their post-industrial phase, but developing countries need much higher rates to converge.
- There are very few resource-rich exceptions to manufacturing-led growth.

Main Conclusions & Key Takeaways

- **The Phenomenon:** Late industrializers reach manufacturing peaks at 40% of pre-1990 income levels.
- **The Causes:** Technology drives it in advanced economies; global trade and falling world prices ($\hat{p}_m < 0$) drive it in developing countries.
- **The Burden:** Low-skilled workers bear the brunt of the job losses.
- **The Threat:** Countries are running out of industrialization opportunities sooner. This blocks the traditional mechanism for rapid growth and catch-up.
- **The future of development will look fundamentally different from the past.**